

Shaani Kamra Lyons

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EDUCATION

Queen's University Kingston, Ontario

Expected April 2028

Bachelor of Applied Science (B.A.Sc) — Electrical Engineering | GPA: 3.74

TECHNICAL SKILLS

Technical: KiCad, AutoCAD, LTSpice, Quartus II, SolidWorks, Arduino Systems, Soldering & Breadboard Testing, Adobe Lightroom

Programming: C, VHDL, Assembly, Arduino Control Logic, Pseudocode

PROJECTS

Custom STM32F411 Development Board

Aug 2026

- Designed a custom microcontroller development board in KiCad, pairing an STM32F411 (Arm Cortex-M4) with an ESP32 Wi-Fi/Bluetooth module.
- Completed a 2-layer PCB layout that passed all electrical and design rule checks on the first submission, then ordered a 5-unit prototype run for hands-on testing.
- Broke out I2C, SPI, and GPIO headers, enabling future expansion with additional sensors without a board redesign.

16-Bit Calculator ALU — FPGA Implementation

Jul 2026

- Designed and implemented a 16-bit calculator ALU in VHDL on an Altera DE0 Nano FPGA, synthesized in Quartus II.
- Built and integrated an adder/subtractor, multiplier, and division state machine into a single working ALU.
- Verified all 4 arithmetic operations produced correct results on hardware after debugging multiple design issues.

Strain Measuring PCB — Queen's Baja SAE

Jan 2025 – Apr 2025

- Designed a custom PCB in KiCad to measure mechanical strain using a Wheatstone bridge and strain gauge, reducing the board footprint by over 90% (7 cm × 7 cm to 3 cm × 2 cm) through iterative design optimization.
- Built and calibrated a signal amplification circuit, confirming a 2x amplification factor through voltage testing.
- Delivered a working prototype on time as part of a 5-person team, coming in \$12 under the \$100 project budget.

Automated Water Pre-Treatment System

Sep 2024 – Dec 2024

- Built an Arduino-controlled prototype that automated water treatment end-to-end, from turbidity sensing through alum dosing to filtration.
- Designed and tested 3 dispensing assembly prototypes in SolidWorks, selecting the design that dispensed 0.7 g of alum on average, within the 0.6–1.0 g target range.
- Achieved consistent turbidity readings of 3.8–4.4 V after treatment, meeting the project's water quality target.

WORK EXPERIENCE

Lakelands Irrigation Barrie, ON

May 2026 – Sep 2026

Irrigation Technician

- Installed, started up, and serviced 300+ residential and commercial irrigation systems, including electrical controllers and wiring.
- Diagnosed and resolved mechanical and electrical system issues across multiple job sites through troubleshooting and service calls.

EngSoc Photography Team Kingston, ON

Oct 2025 – Oct 2026

Photographer

- Photographed engineering events, competitions, and conferences for EngSoc, managing shoots independently through final delivery.

AWARDS AND ACCOLADES

- Dean's Scholar
- Sir Isaac Newton Exam Distinction — University of Waterloo
- Ontario Scholar